Analytical Problems

<u>Problem: 1.</u> Assume the following regarding the development of a software system P:

- Estimated lines of code of P: 33,480 LOC

- Average productivity for P: 620 LOC per person-month

- Number of software developers: 6

- Average salary of a software developer: 50,000 per month.

If E, D, and C are the estimated development effort (in person-months), estimated development time (in months), and estimated development cost (in Lac) respectively, then (E, D, C).

Solution:

1. Estimating Development Effort (E) in Person-Months:

We use the formula:

E = Total Lines of Code (LOC) / Productivity (LOC per person-month)

Given:

- Total LOC = 33,480

- Productivity = 620 LOC per person-month

E = 33,480 / 620 = 54 person-months

2. Estimating Development Time (D) in Months:

We use the formula:

D = E / Number of Developers

Given:

- E = 54 person-months

- Number of developers = 6

D = 54 / 6 = 9 months

3. Estimating Development Cost (C) in Lac:

We use the formula:

C = D * Average Salary per Developer per Month

Given:

- D = 9 months

- Salary per developer per month = 50,000

Total cost for one developer:

C(one developer) = 9 * 50,000 = 450,000

For 6 developers:

C = 6 * 450,000 = 2,700,000 = 27 Lac

Problem:2. A software project was estimated at 864 Function Points. A six-person team will be assigned to the project consisting of a requirement gathering person, one designer, two programmers, and two testers. The salary of the designer is 70,000 per month, the requirement gatherer is 50,000 per month, the programmer is 60,000 per month, and a tester is 60,000 per month. Average productivity for the team is 12 FP per person-month. Calculate the projected cost of the project?

Solution:

1. Estimating the Development Effort (E) in Person-Months:

We use the formula:

E = Total Function Points (FP) / Productivity (FP per person-month)

Given:

- Total FP = 864

- Productivity = 12 FP per person-month

E = 864 / 12 = 72 person-months

2. Estimating the Development Time (D) in Months:

The team consists of 6 people, so the time required is:

D = E / Number of Developers

D = 72 / 6 = 12 months

3. Estimating the Monthly Cost for the Team:

The team consists of the following members with corresponding salaries:

- Requirement Gatherer: 50,000 per month

- Designer: 70,000 per month

- Two Programmers: 60,000 per month (each)

- Two Testers: 60,000 per month (each)

Total Monthly Salary = 50,000 + 70,000 + 2(60,000) + 2(60,000) = 360,000 per month

4. Estimating the Total Cost (C):

The total cost for the project is the monthly cost multiplied by the number of months (D):

C = D * Total Monthly Salary

Given:

- D = 12 months

- Total Monthly Salary = 360,000

C = 12 * 360,000 = 4,320,000 = 43.2 Lac

Problem:3.

A company has a choice of two languages L1 and L2 to develop a software for their client. The number of LOC required to develop an application in L2 is thrice the LOC in language L1. Also, the software has to be maintained for the next 10 years. Various parameters for the two languages are given below to decide which language should be preferred for development:

Parameter	L_1	L_2
Man-year needed for development	LOC/1000	LOC/1000
Development cost	Rs. 70,000	Rs. 90,000
Cost of Maintenance per year	Rs. 1,00,000	Rs. 40,000

The total cost of the project includes the cost of development and maintenance. What is LOC for L1 for which the cost of developing the software with both languages must be the same?

Solution:

The problem is solved as follows:

Given:

- LOC required in L2 is thrice the LOC in L1.

- Development cost and maintenance cost are provided for both languages.
- The software is to be maintained for 10 years.

Total development and maintenance costs were equated, leading to the following equation:

(LOC / 1000) * 70000 + 1000000 = (3 * LOC / 1000) * 90000 + 400000

So, LOC = 3000.

Problem:4. A software company needs to develop a project that is estimated as 1000 function points and is planning to use JAVA as the programming language whose approximate lines of code per function point is accepted as 50. Considering a = 1.4 as multiplicative factor, b = 1.0 as exponentiation factor for the basic COCOMO effort equation and c = 3.0 as multiplicative factor, d = 0.33 as exponentiation factor for the basic COCOMO duration equation, approximately how long does the project take to complete?

Solution:

Problem:5. The number of function points of a proposed system is calculated as 500. Suppose that the system is planned to be developed in Java and the LOC/FP ratio of Java is 50. Estimate the effort (E) required to complete the project using the effort formula of basic COCOMO. Assume that the values of a and b are 2.5 and 1.0 respectively.

Solution:

Problem.6: "A new project with an estimated 400 KLOC embedded system has to be developed. The Project Manager has a choice of hiring from 2 pools of developers:

- Very highly capable (with appropriate skills) but with very little experience in programming language, or
- Developers of low quality but a lot of programming language experience.

Which is the better choice in terms of the 2 pools?"

Solution:

Problem:7. Using the Intermediate COCOMO model, estimate the effort required to develop the enterprise resource planning (ERP) system which is semidetached project based on the following project characteristics:

Size of the software: The estimated size of the ERP system is 100,000 lines of code.

Product attributes: The project is considered to have 'nominal' requirements, meaning that it involves some complex processing but falls within the typical range of complexity for similar ERP systems.

Personnel attributes: Experienced software engineers with good knowledge might reduce the effort by 1.10 (for skilled personnel).

Project attributes: Since the project schedule is 'tight' with moderate risk, we'll increase the effort by 1.15 (representing a tight schedule with moderate risk).

Based on these characteristics, identify the effort required in person-months to develop the ERP system using the Intermediate COCOMO model. Additionally, identify the estimation of the project duration in months.

Solution:

Intermediate COCOMO Model Problem Solution:

Given Data:

- Size of the ERP system: 100,000 lines of code (LOC)

- Project type: Semidetached

- Effort Adjustment Factors (EAF):

  - Personnel attributes: Skillful personnel reduce effort by a factor of 1.10.

  - Project attributes: Tight schedule increases effort by a factor of 1.15.

Intermediate COCOMO Effort Equation:

Effort (E) in person-months is given by the formula:

$$E = a * (KLOC)^b * EAF$$

Where:

- E = Effort in person-months

- a, b = Constants based on project type (Semidetached)

- KLOC = Size of project in thousands of lines of code = LOC / 1000

- EAF = Effort Adjustment Factor (product of all applicable cost drivers)

Constants for a Semidetached Project:

- a = 3.0

- b = 1.12

Step 1: Calculate KLOC

KLOC = 100,000 LOC / 1000 = 100 KLOC

Step 2: Calculate EAF

EAF = 1.10 * 1.15 = 1.265

Step 3: Calculate Effort in Person-Months

E = 3.0 * (100)^1.12 * 1.265

E ≈ 3.0 * 132.2 * 1.265 ≈ 501.4 person-months

Step 4: Estimate Project Duration in Months

Duration (T) in months is given by the formula:

T = 2.5 * (E)^0.38

T = 2.5 * (501.4)^0.38

T ≈ 2.5 * 13.1 = 32.8 months